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Mitigation of Broadcast Storm Problem for Reliable Emergency Dissemination in IoV using Trust-Aware Clustering and Batch Authentication

23CSE399 – Project Phase 1 (Panel Review 1)

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1 Problem Definition & Originality

The Internet of Vehicles (IoV) enables real-time communication among vehicles for safety-critical applications such as accident alerts and traffic management. However, in dense traffic environments, emergency message dissemination suffers from the **broadcast storm problem**, where multiple vehicles simultaneously rebroadcast alerts.

This leads to channel congestion, packet collisions, increased latency, and reduced reliability. The system must ensure ultra-low latency for time-critical safety applications.

Real-world impact:

- Highway pile-ups delay alerts
- Urban congestion blocks communication
- Malicious nodes inject false alerts

Our Contribution:

- Trust-aware clustering
- Controlled forwarding
- RSU batch authentication

2 Literature Survey & Gap Analysis

Trust-based dissemination improves reliability [1]. Hybrid trust models enhance robustness [2]. Cluster-based approaches improve scalability [3]. Time-barrier methods reduce redundancy [4]. RSU-assisted methods improve latency [5].

Limitations:

- Uncontrolled broadcast flooding
- No trust-based CH selection
- High authentication overhead

Research Gap:

- No unified integration of trust, clustering, and authentication

3 Comparative Analysis of Existing Work

Year	Approach	Advantage	Limitation
2024	TCEMD	High trust accuracy	No clustering
2024	HTEMD	Robust to malicious nodes	No broadcast control
2024	VEMD	Reduces packet loss	No trust model
2019	Time Barrier	Reduces rebroadcasts	Fails in dense traffic
2023	RA-HEMB	Low latency	No authentication filtering

Table 1: Comparison of Existing Methods

4 Research Gap vs Proposed Solution

Problem	Limitation	Solution
Broadcast storm	Multiple rebroadcasts	CH forwarding
Trust issues	Malicious nodes	Trust-based CH selection
Authentication delay	Per-message verification	Batch authentication

Table 2: Gap Mapping

5 System Architecture

The proposed architecture is designed to ensure efficient and secure emergency message dissemination in IoV environments.

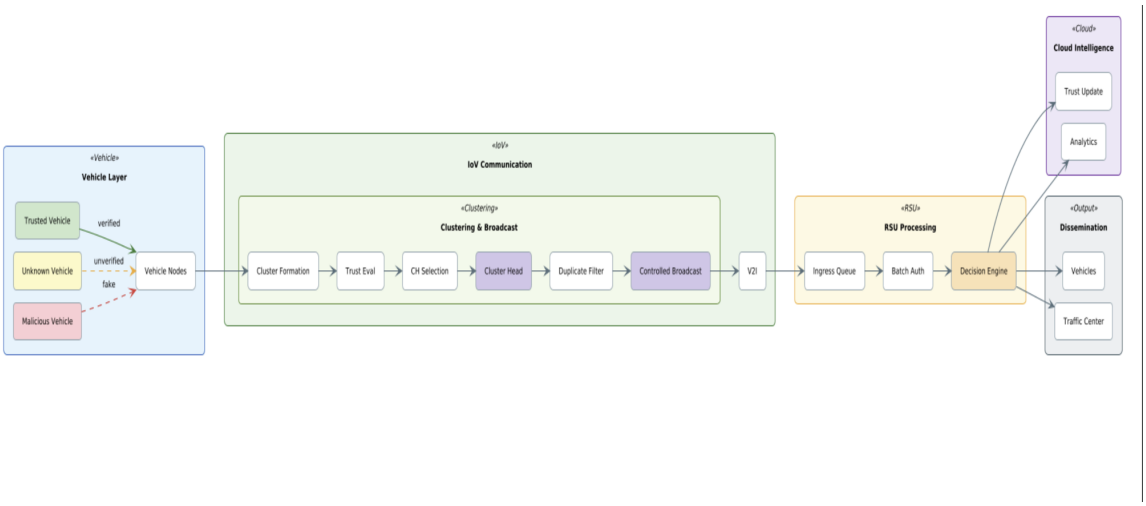

Figure 1: Proposed IoV Emergency Message Dissemination Architecture

Layers:

- IoV Communication Layer
- Clustering Layer (DBSCAN)
- Trust Evaluation Layer
- Cluster Head Selection
- Controlled Forwarding
- RSU Batch Authentication

6 Proposed Methodology

System Flow:

IoV → Clustering → Trust → CH Selection → Controlled Forwarding → RSU Authentication

DBSCAN:

$$N_\epsilon(p) = \{q \mid dist(p, q) \leq \epsilon\}$$

Time Complexity: $O(n \log n)$

Trust Model:

$$T = w_1 T_{direct} + w_2 T_{historical} + w_3 T_{feedback}$$

Batch Authentication:

$$e\left(\sum S_i, g\right) = \prod e(H(M_i), PK_i)$$

7 Algorithm Summary

Algorithm	Layer	Purpose
DBSCAN	Clustering	Dynamic grouping of vehicles
Trust Model	CH Selection	Evaluate node reliability
BLS	RSU Layer	Batch message verification
Kalman Filter	Analysis	Traffic prediction

Table 3: Algorithms Used

8 Detailed System Workflow

- Event detection using sensors
- Initial broadcast
- Clustering
- Trust evaluation
- Cluster head selection
- Controlled forwarding
- RSU authentication
- Monitoring

9 Security Analysis

Trust filtering prevents malicious nodes [6]. Batch authentication ensures secure and scalable communication.

10 Performance Evaluation Strategy

- Latency
- Packet Delivery Ratio
- Network overhead
- Authentication delay

11 Future Enhancements

- 5G V2X
- AI-based trust
- Blockchain security

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